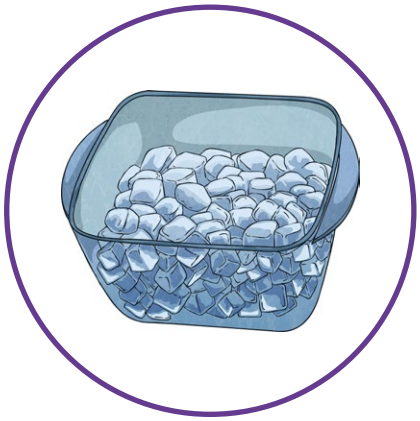
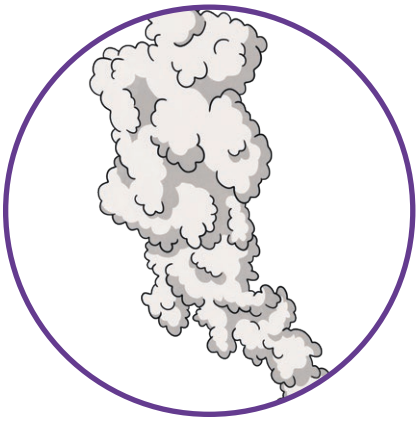
SCIENCE

States of Matter

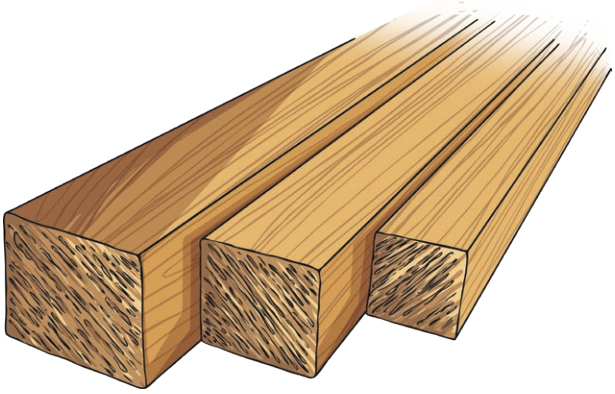
All matter is made up of atoms, but did you know there are three common states of matter? They are solid, liquid, and gas.







Atoms in a solid state of matter are closely packed together. In fact, they are so tightly packed that they really cannot move, only vibrate. Solids have a definite shape and volume. This means the shape and volume do not change. Some examples of solids are a piece of wood, your family's computer, your favourite car, and an ice cube.

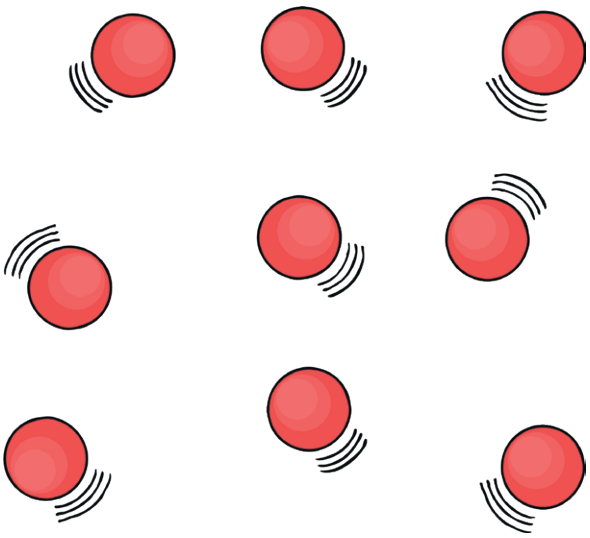


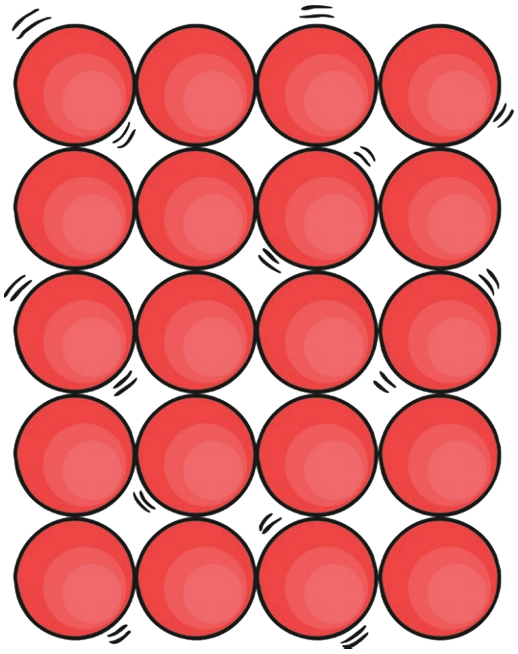


Atoms in a liquid are not as closely packed together as a solid. They are in an organised order but can move a little bit. Liquids have a definite volume but take the shape of the container they are in. For example, if you poured a cup of water into a cylinder or cube, the water will take the shape of the cylinder or cube. Regardless of the container, its volume (one cup) will remain the same. Some examples of liquids are water, oil and juice.

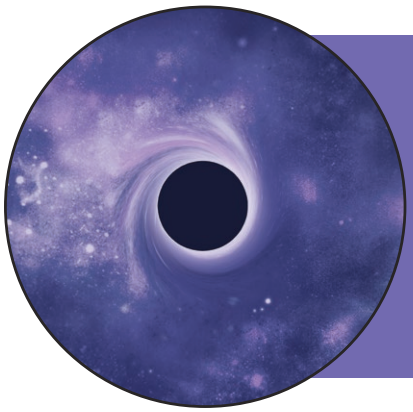
States of Matter

Atoms in a gas move freely. They are not in an organised arrangement and have random motion. They have an indefinite volume and shape. This means their volume and shape change depending on where they are. For example, if you put steam into a big soup pot or into a box, the steam will spread out in each container to fill the volume and shape of the container. Some examples of gases are water vapour, oxygen, and nitrogen.





Did you know there is actually a fourth state of matter? It is called plasma. Plasma is the most common state of matter in the universe (but not very common on Earth). Atoms in plasma move very fast (have lots of kinetic energy), and their electrons group together so the atoms act as one instead of different parts. Plasma is present in stars, fluorescent lights, and even some televisions.



There is also a fifth state of matter called Bose-Einstein condensates (BEC). BEC matter joins all molecules together to create a "super-molecule." Though not very common, scientists believe BEC matter is found in black holes.

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States of Matter Questions

1. What is the most common state of matter in the universe?

   ○ solid ○ gas

   ○ liquid ○ plasma

2. Which state of matter is thought to be present in black holes?

   ○ liquid ○ plasma

   ○ Bose-Einstein condensates ○ gas

3. Which state of matter has an indefinite volume and shape?

   ○ gas ○ liquid

   ○ solid ○ plasma

4. What makes each state of matter different?

   ○ amount of atoms ○ size of atoms

   ○ movement (energy) of atoms ○ shape of atoms

5. What states of matter were present in your breakfast this morning? (List the item and its state of matter)

6. Can the same molecules be changed from one state of matter to another? Provide at least one example.